Material Characterization and Instrumentation Techniques

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Outline

1. X-ray Diffractometer
   - Scherrer equation
2. X-ray Photoelectron Spectroscopy (XPS)
3. Scanning Electron Microscope (SEM)
4. Atomic Force Microscopy (AFM)
5. Transmission Electron Microscopy (TEM)

X-ray Diffractometer

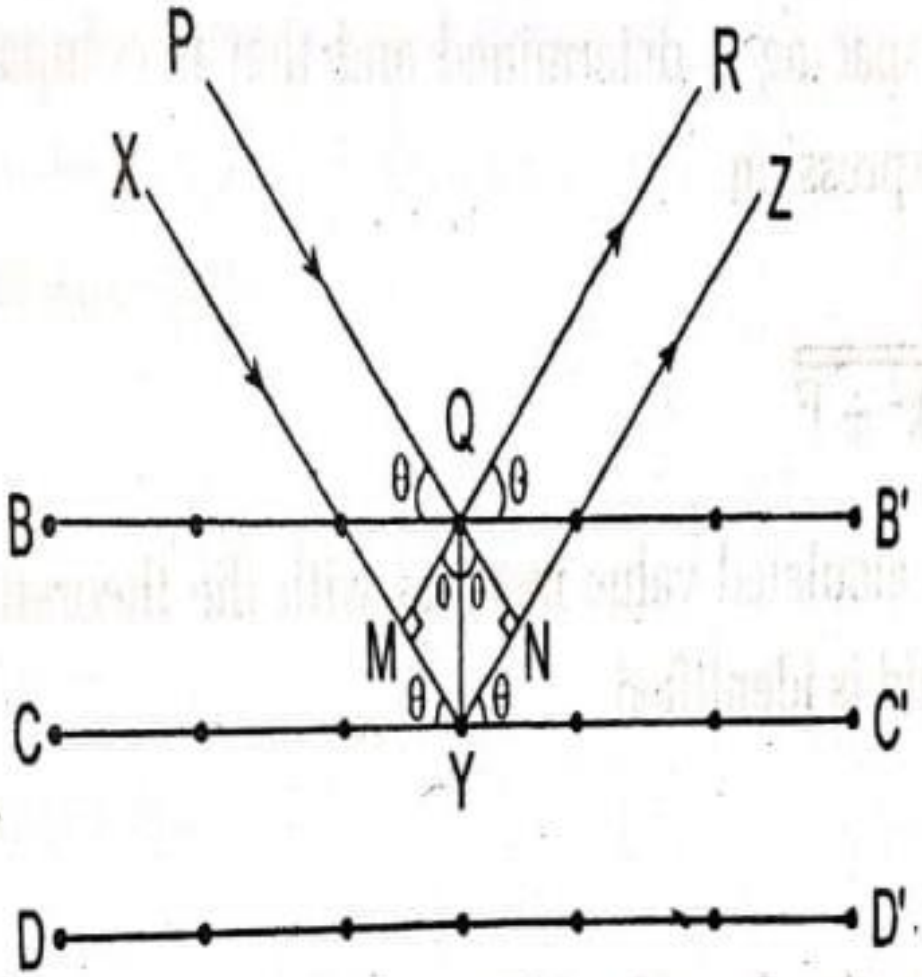
- Gives inter-planar distance in a crystal.
- Can provide information on unit cell dimensions.

Principle:

- X-ray diffraction is based on interference of monochromatic X-rays reflected from a crystalline sample.
- These X-rays are generated by a cathode ray tube.
- Further filtered to produce monochromatic radiation, collimated to concentrate, and directed towards the sample.
- It produces constructive interference when conditions satisfy Bragg's Law ($n\lambda = 2d \sin\theta$). This law relates the wavelength of electromagnetic radiation to the diffraction angle and the lattice spacing in a crystalline sample.

- These diffracted X-rays are then detected, processed and counted. By scanning the sample through a range of 2θ angles, all possible diffraction directions of the lattice should be attained due to the random orientation of the powdered material.
- Conversion of the diffraction peaks to lattice spacings allows identification of the mineral because each mineral has a set of unique lattice spacings.
- Typically, this is achieved by comparison of lattice spacings with standard reference patterns.

Derivation of Bragg's equation

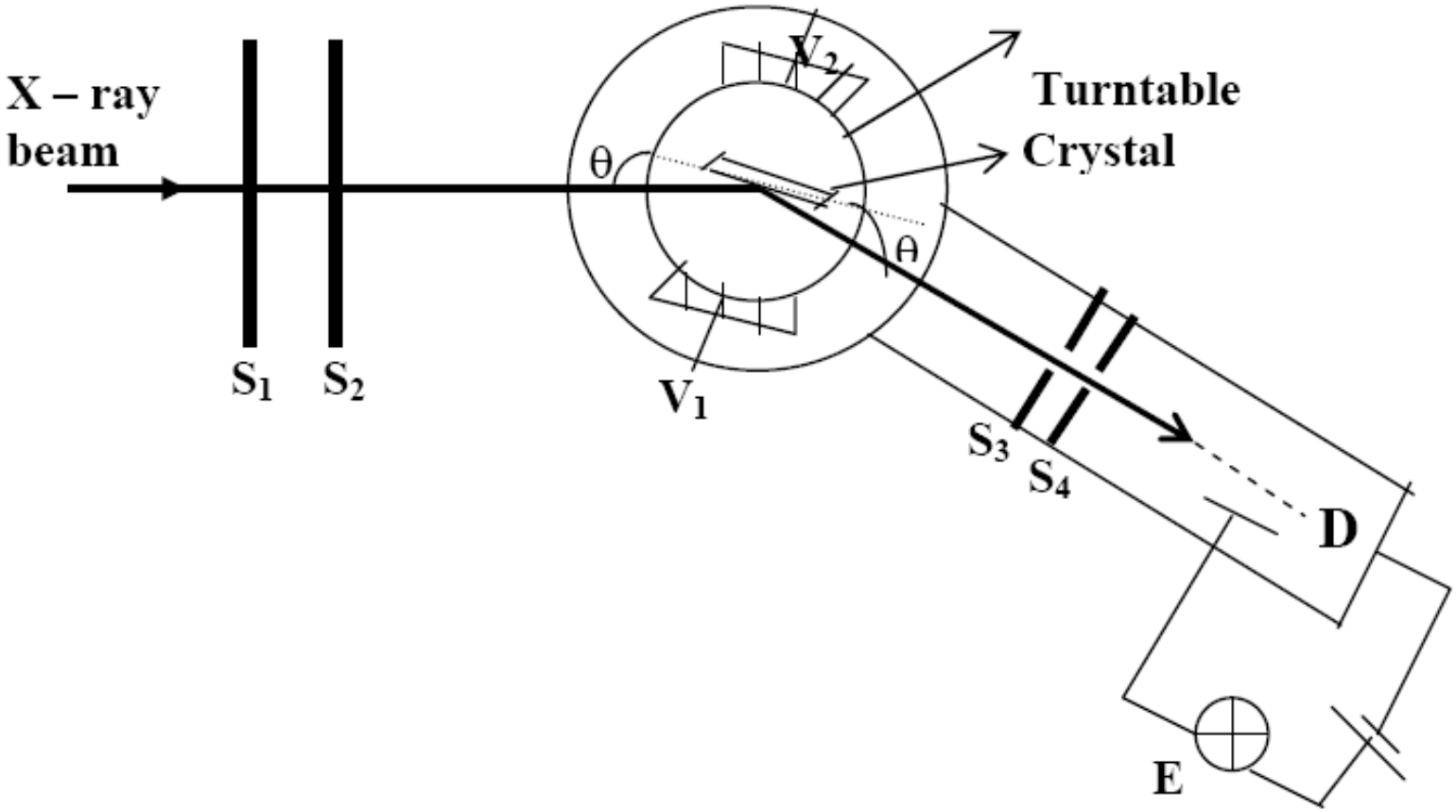


- Consider a crystal's parallel planes with the inter planar distance d.
- Consider a beam of monochromatic X-rays PQ incident on plane BB′ with grazing angle θ and reflected along QR.
- Similarly, another beam XY is incident on the second plane CC′ with angle θ and reflected along YZ.
- MY = NY = $d \sin\theta$
- The path difference between the two reflected rays = MY+NY $= d \sin\theta + d \sin\theta = 2d \sin\theta$.
- Now, in order for reflected rays to be in phase (for constructive interference), the path difference between them must be an *integral multiple* of the X-ray wavelength (λ),

$$n\lambda = 2d \sin\theta \qquad \text{with} \qquad n = 1, 2, 3, \ldots$$

- This equation is called Bragg's equation.

Construction and working of Diffractometer

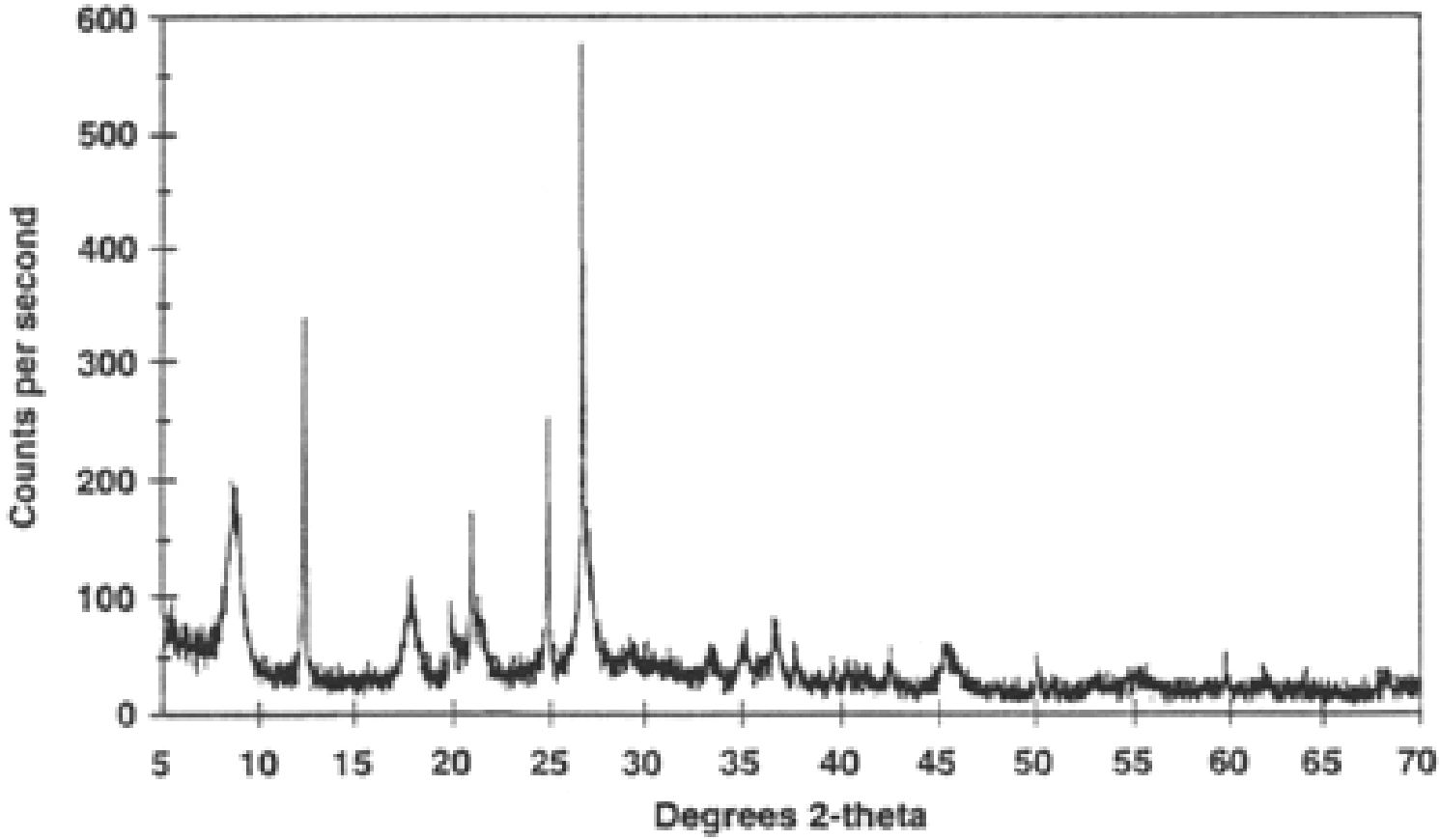


- X–rays are produced from an X–ray tube by bremsstrahlung and Kα characteristic X-rays.
- Slits S_1 and S_2 makes the beam narrow.
- Beam is made to fall on the crystal C fixed at the center of a circular turn table.
- The X – ray beam after reflection, enters an ionization chamber D.
- The ionization chamber is mounted on a mechanical arm which can turn coaxially with the turn table.
- But the mechanical arm is so coupled to the turn table, for every rotation by an angle θ of the turn table, the mechanical arm turns to an angle of 2θ.

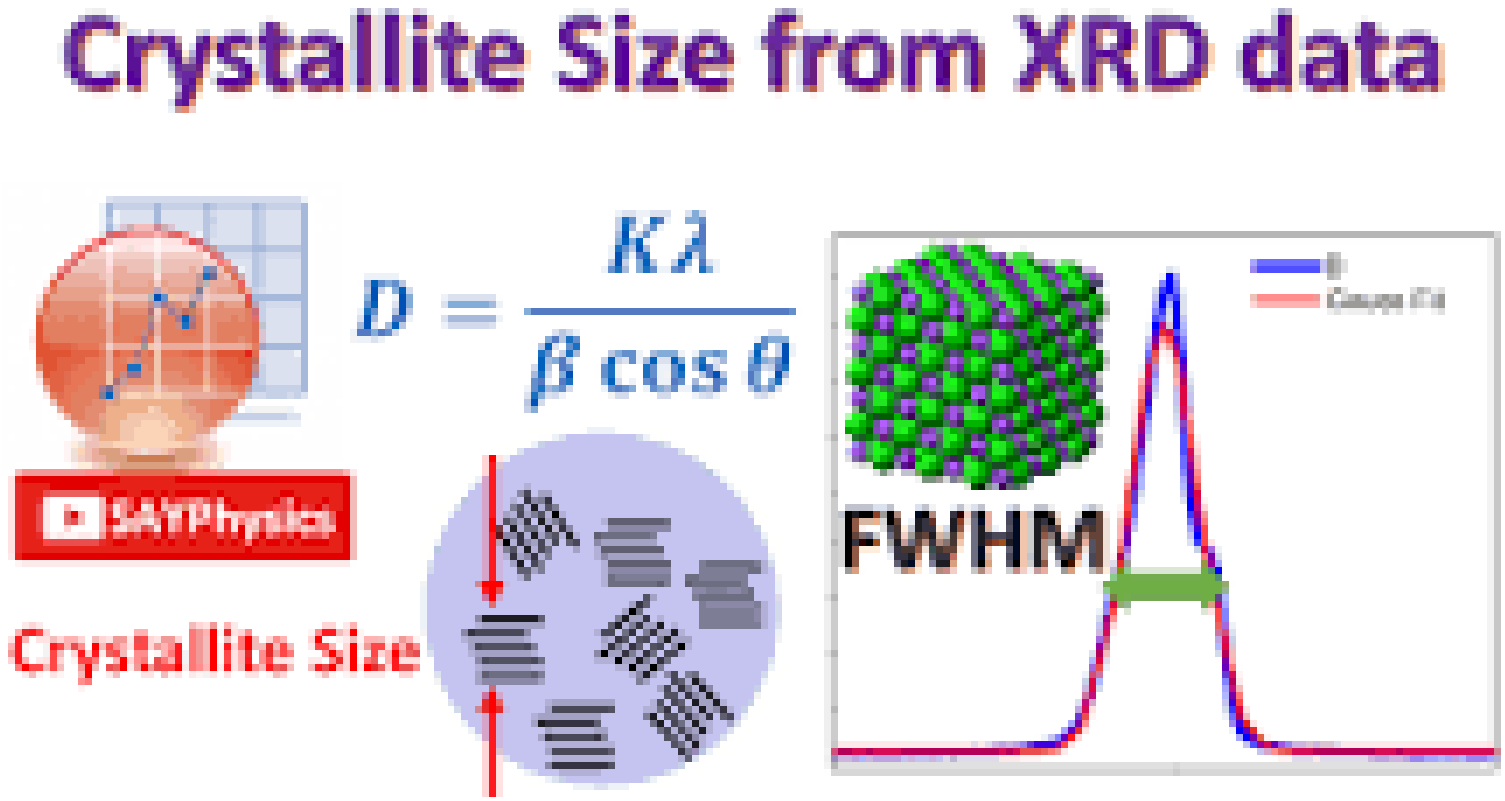
The Bragg's spectrometer is used to:

- Verify Bragg's law
- Determine the wavelength of monochromatic x-ray
- Analysis of crystal structure

Copper is the most common target material used to produce X-rays for single-crystal diffraction, with wavelength of CuKα radiation = 1.5418Å.


Counts per second
600
500
400
300
200
100
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
Degrees 2-theta

Crystallite size determination by Scherrer equation



Crystallite is a small region of a solid that consists of a single crystal; a grain.

- The crystallite size can be calculated from the full width at half maximum intensity (FWHM) measured on the corrected diffraction profile.
- The crystal size can be determined by using Scherrer Equation which is given by

$$D = \frac{k\lambda}{\beta \cos\theta}$$

- Where λ is the wavelength of X-rays, β is the full width at half maximum of peaks obtained at an angle 2θ. Shape factor of k can be between 0.62–2.08 and is usually taken as about 0.89.

X-ray Photoelectron Spectroscopy (XPS)

Electron spectroscopy for chemical analysis (ESCA) is primarily used for identifying the chemical species present in the film.

Principle:

- Incident X-rays cause photoemission of electrons from the sample surface.
- These electrons energy is analyzed to obtain the binding energy of the emitted electrons.
- When X-ray photons are incident on a solid, they eject electrons (photoelectrons) from the valence band.
- Electrons can be emitted from any orbital, and photoelectron emission occurs as long as the X-ray energy exceeds the binding energy of the electrons.

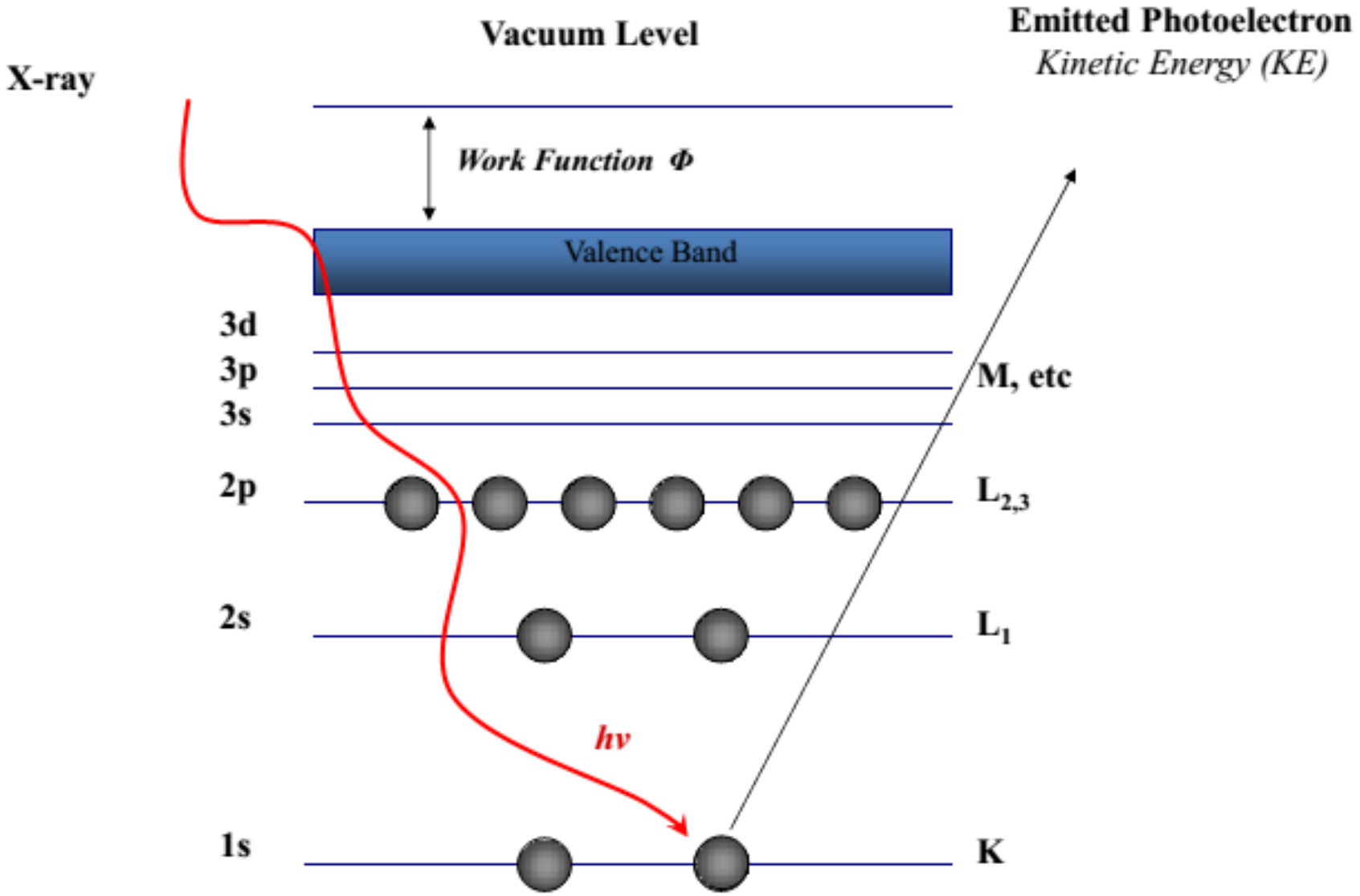
- The determination of the kinetic energy of the outgoing electron is the cornerstone of XPS.

$$E_K = h\nu - E_B - \phi$$

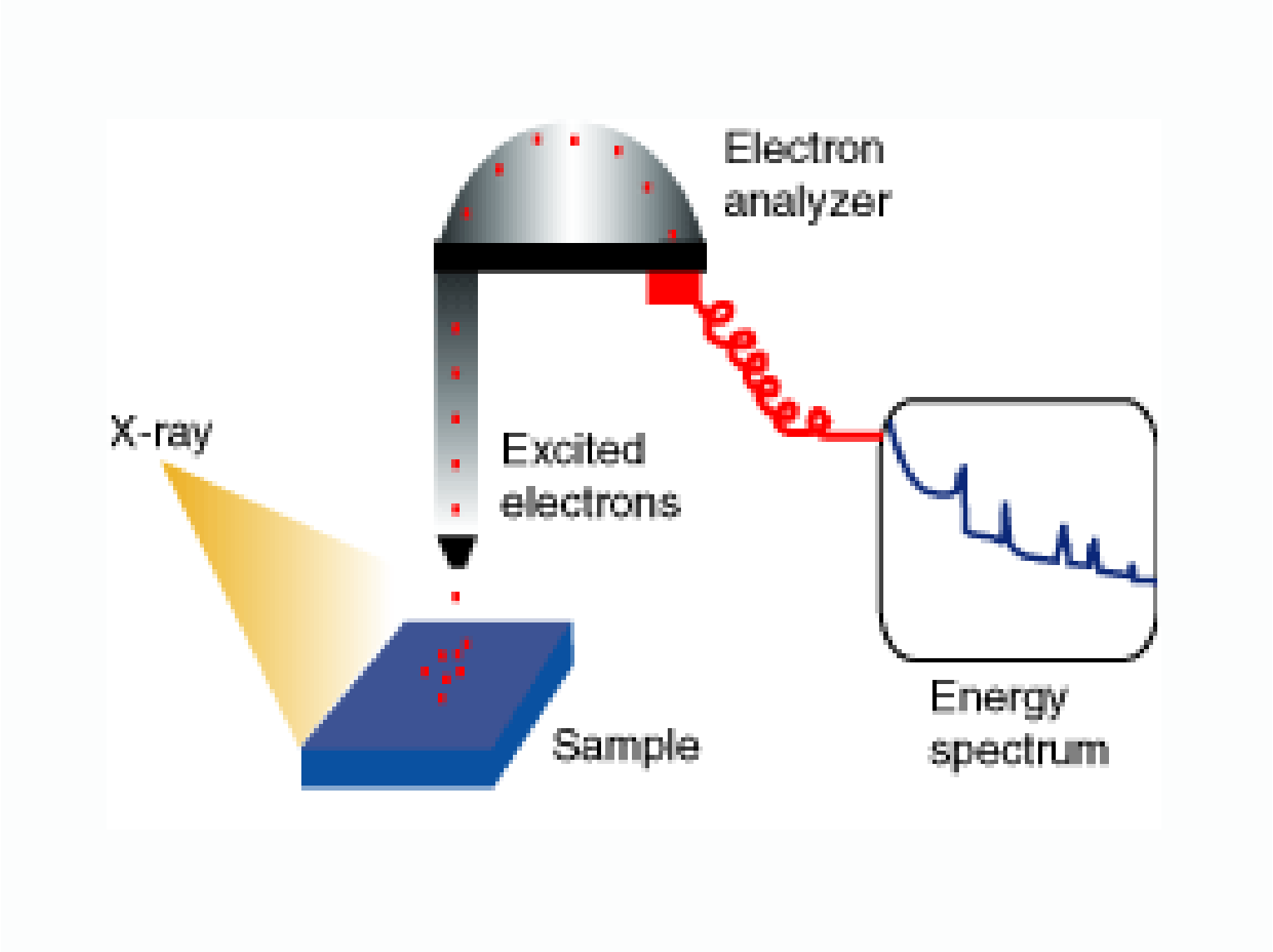
where $h\nu$ is the energy of the incident X-ray photon, E_B is the binding energy of the electron and ϕ is the work function.

- The electron binding energy E_B is influenced by its chemical surroundings, and is suitable for determining the chemical states of the atoms by measuring the energy shifts.

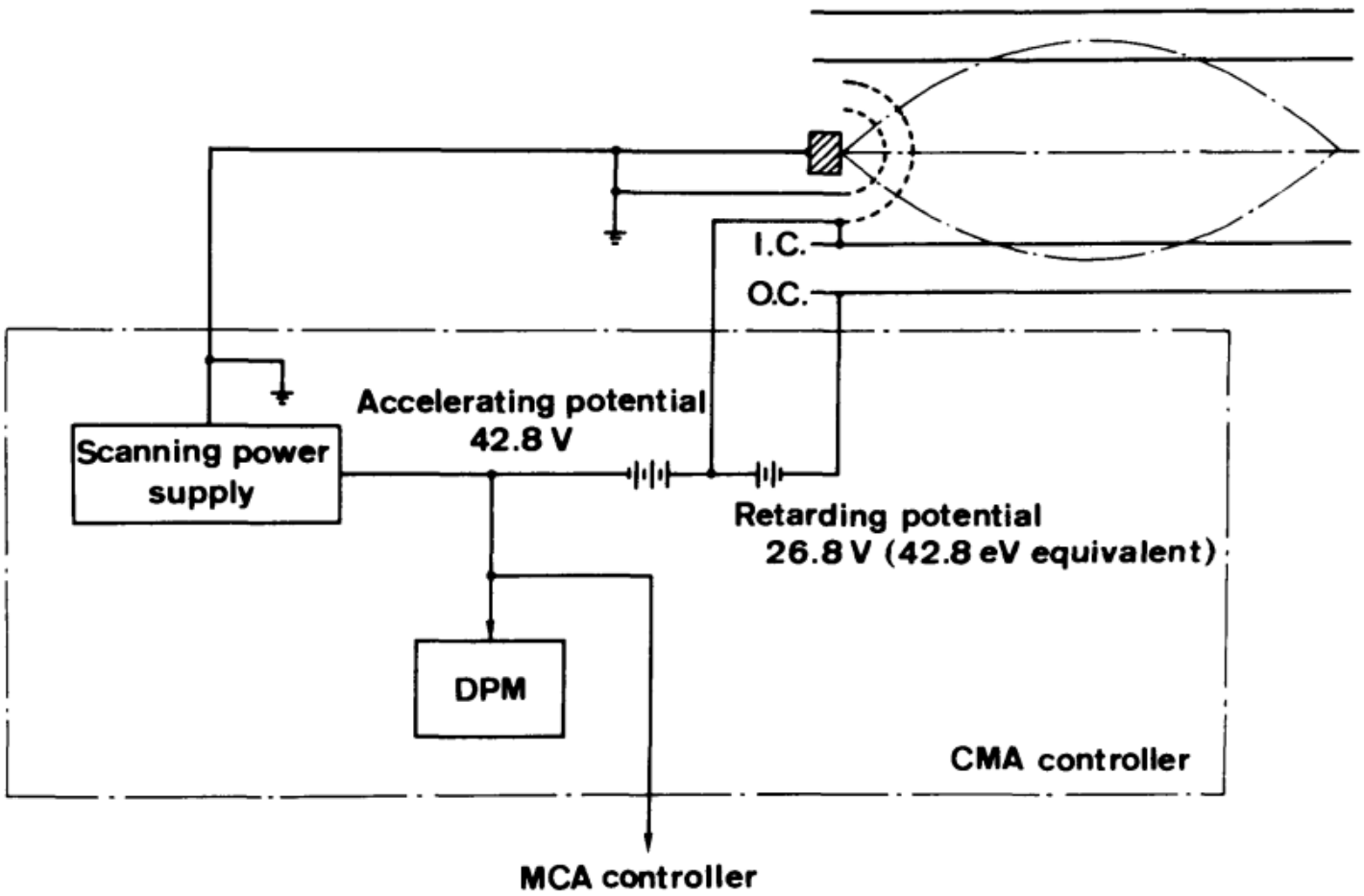
X-ray Photoelectron Spectroscopy (XPS)



Construction:



- X-ray source:
  - X-ray energy must be high enough to knock out the electrons.
  - X-rays should be as monochromatic as possible. Typically the $K\alpha$ characteristic X-rays from Magnesium (1253.6 eV) or Aluminum (1486.6 eV) is used for analysis.
- The XPS electrons are detected with the most common Cylindrical Mirror Analyzer (CMA). A "double-pass CMA" is generally used to discriminate against small energy shifts.
- Chemical compounds or elements are identified by the location of the energy peaks in the photoelectron spectrum.
- Vacuum should be maintained so that the emitted electrons are not slowed down due to collision with surrounding gas atoms.



Cylindrical Mirror Analyzer (CMA)

Scanning Electron Microscope (SEM)

- The SEM uses electrons instead of light to form an image.
- Used for imaging the surfaces of almost any material with a resolution down to about 1 nm.
- Have magnification ranges from 20X to 100000X.

Principle:

- The smallest distance resolved by a microscope is called the limit of resolution (Δx) and the resolving power of a microscope is the reciprocal of the limit of resolution.

$$\Delta x = \frac{\lambda}{2\mu \sin\theta} = \frac{\lambda}{\text{NA}} \qquad \text{Resolving Power} = \frac{1}{\Delta x} = \frac{\text{NA}}{\lambda}$$

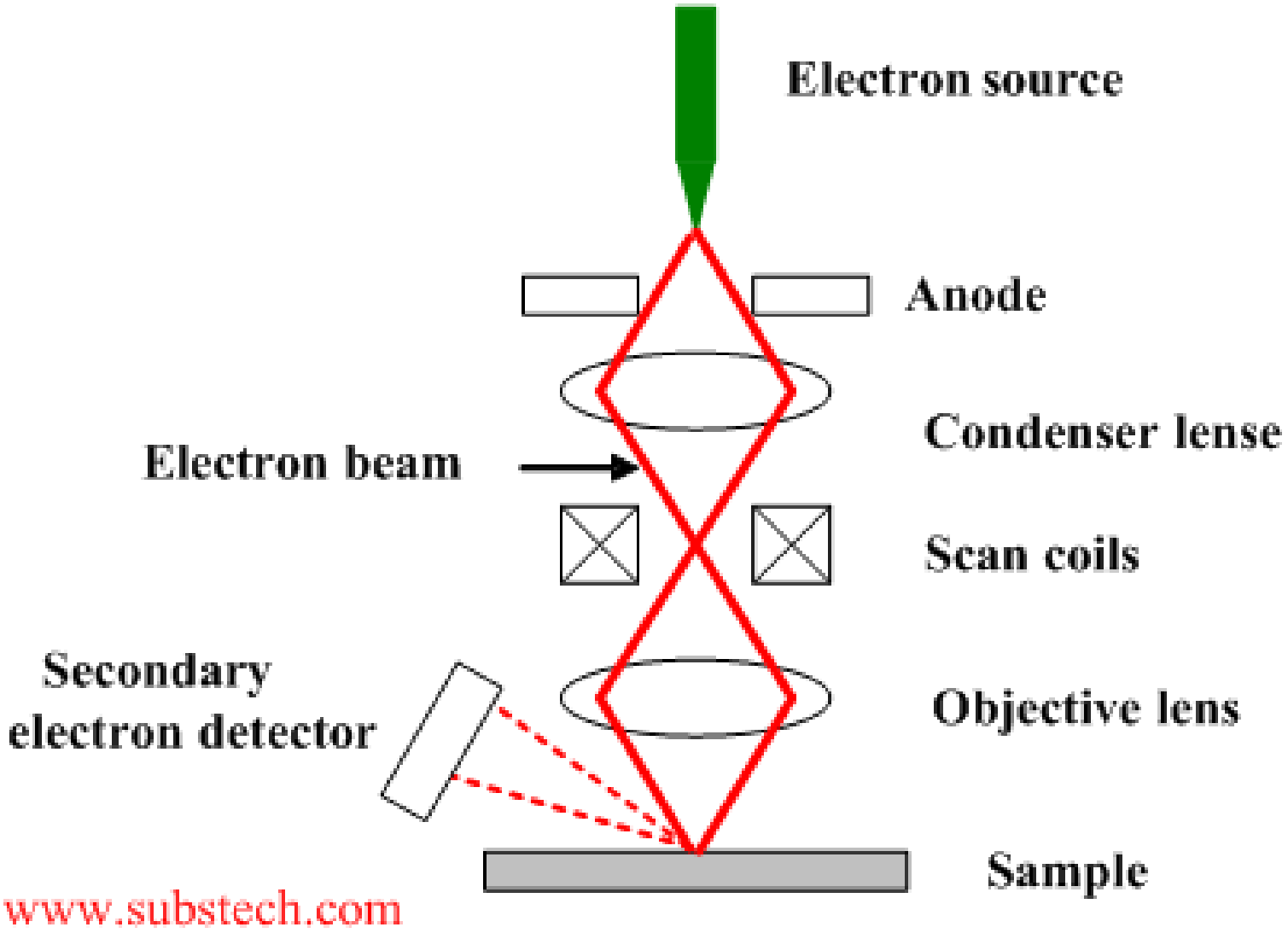
where λ is the wavelength of the light used, μ is the refractive index of the medium separating object and aperture. θ is the half-angle subtended by the first minimum at the source. NA $= 2\mu \sin\theta$ is clled the numerical aperture.

- Thus shorter the wavelength of the radiation, greater is the resolving power.
- A mere increase in the magnifying power without a corresponding increase in the resolving power would only produce large blurred image.
- Optical microscope uses a light beam of wavelength ≈ 600 nm so that the smallest distance resolvable ≈ 300 nm (Assuming numerical aperture $= 2$).
- Hence, if instead of light waves we could use electron beam then the resolving power can be increased by more than 10^6 fold.
- de Broglie wavelength of electrons accelerated by a voltage of V is given by,

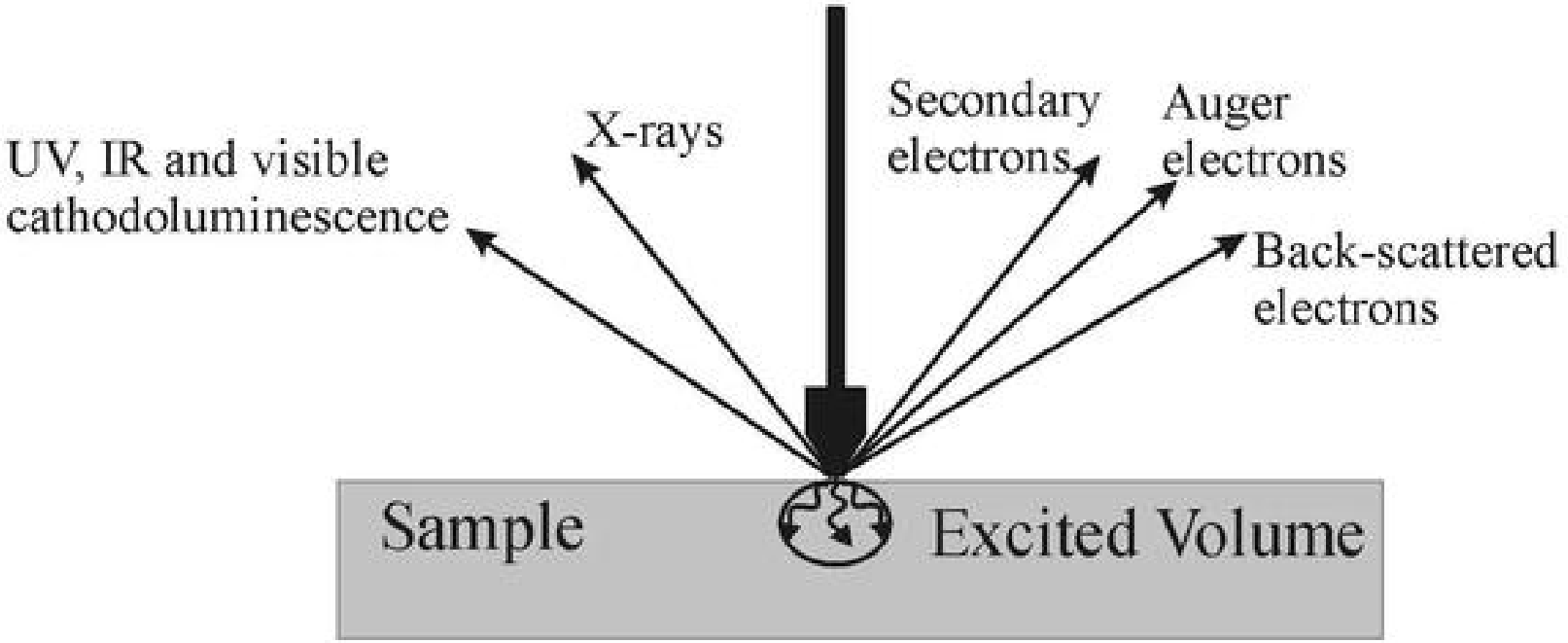
$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE_k}} = \frac{h}{\sqrt{2meV}} \qquad \text{since } E_k = \frac{p^2}{2m}$$

- For electrons accelerated to approximately 75 kV will have wavelength of 4.48×10^{-12} m which is roughly 10^5 times shorter than visible radiation.

Construction and working:



- The magnetic fields which are axially symmetric have a focusing effect on the electron beam passing through them.
- The axially symmetric magnetic fields are produced by short solenoids. Such solenoids are called magnetic lenses.
- By adjusting the current through the coils the intensity of the magnetic field and hence the focal length can be adjusted.
- Electrons are thermionically emitted from a tungsten filament (cathode) are drawn to an anode (with an acceleration voltage of 1 KV to 40 KV) and focused by two successive magnetic lenses into a beam with a very fine spot size that is typically 10Å in diameter.
- These electrons interact with the sample to depth of approximately 1µm.
- Pairs of scanning coils located at the objective (magnetic) lens deflect the beam either linearly or in raster fashion over a rectangular area of the specimen surface.


UV, IR and visible
cathodoluminescence
X-rays
Secondary
electrons
Auger
electrons
Back-scattered
electrons
Sample
Excited Volume

- Upon impinging on the specimen, the primary electrons decelerate and several processes such as elastic scattering viz.,
  - forward scattering and
  - backscattering of the incoming electrons

  and inelastic scattering viz.,
  - generation of secondary electrons,
  - Auger electrons,
  - Bremsstrahlung,
  - characteristic X-rays, etc.
- Various SEM techniques are differentiated on the basis of what is subsequently detected and imaged, and the principle images produced in the SEM are of three types:
  - Secondary electron images,
  - Backscattered electron images and
  - Elemental X-ray maps.

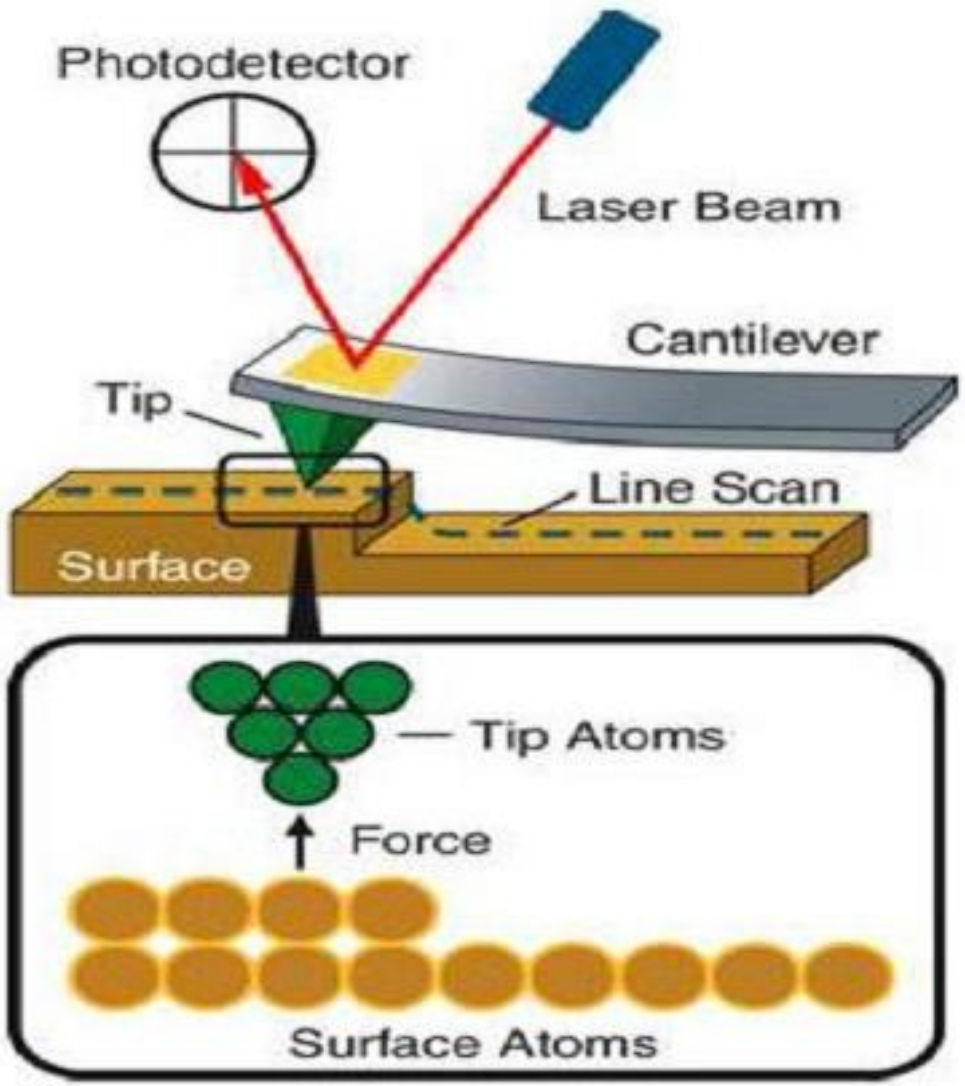
- The number of electrons produced at any given point can be related directly to the topography of the specimen with respect to the detector.
- The patterns created on the viewing screen represent the surface topography of the specimen.
- The probability of backscattering increases with the atomic number of the sample material.
  - Therefore the primary electrons arriving at a given detector position can be used to yield images containing information on both topology and atomic composition.
- An additional electron interaction in the SEM is that the primary electron collides with and ejects a core electron from an atom in the sample.
  - The excited atom will decay to its ground state by emitting either a characteristic X-ray photon or an Auger electron.
  - By analyzing energies of characteristic of the x-ray photon the type of atoms can be identified.
  - Further the concentration of atoms in the specimen can be determined by counting the number of X rays emitted.

Atomic Force Microscopy (AFM)

Principle:

- Atomic Force Microscope (AFM) is a high-resolution microscopy technique which produces precise 3-dimensional topographic images of a sample by scanning the surface with a nanometer-scale probe/tip (lateral resolution $\approx 1\,\text{nm}$, vertical $\approx 0.1\,\text{nm}$) and by measuring attractive or repulsive forces between a tip and the sample.
- These deflection are used to study the morphological and textural characteristics like roughness, waviness, lay and flaws of diverse surfaces along with the physical properties like elasticity, adhesion, hardness and friction.

Construction and working:



- A cantilever with a sharp tip is scanned across the surface of the specimen. The short range or long range forces between the tip of the cantilever and the specimen surface cause the deflection of the cantilever according to Hooke's law.
- This deflection of the cantilever is measured using the change in the reflection angle of the laser spot incident on the top surface of the cantilever which behaves like a mirror.
- This reflected laser light is collected by a position sensitive detector to generate a map of the surface topography.
- There are three main operational modes viz.
  1. Contact AFM (Probe-surface separation is $<0.5\,\text{nm}$)
  2. Intermittent contact/tapping mode AFM (Probe-surface separation is $0.5 - 2\,\text{nm}$)
  3. Non-contact AFM (Probe-surface separation range is $0.1 - 10\,\text{nm}$)

- In contact mode, the cantilever drags across the sample surface and it uses the deflection of the cantilever to measure the contours of the surface. To eliminate the noise and drift that can affect a static signal, low stiffness cantilevers are used.
- In non-contact mode, the tip vibrates slightly above its resonance frequency and does not contact the surface of the sample. Any long range forces, like van der Waals forces, decreases the resonant frequency of the cantilever.
- In tapping mode, the cantilever uses a piezoelectric element mounted on the top to oscillate it at near to its resonance frequency with amplitude of up to 200nm. The forces cause the amplitude to decrease as the tip gets close to the surface.
- This tapping results in less damage to the sample than contact mode and is more accurate than non-contact mode when moisture is present on a sample.

Cantilever and the tip:

- The stiffness of the caintilever needs to be less i.e., the effective spring constant holding atoms together must be in the order of 1 – 10 nN/nm.
- The tip should have a radius of curvature less than 20-50 nm (smaller the better) with a cone angle between 10-20 degrees.
- To obtain both low spring constant and high frequency (>2 KHz), mass of the cantilever should be very small ($\approx 10^{-10}$kg).
- Microlithography process is used to make the cantilevers.

- Si or Si_3N_4 are used for making tips. A thin gold layer is deposited on upper side of the tip for good reflectivity of laser beam.
- Tips are generally evaluated by their "end radius" which limits the resolution of AFM.
- The 'normal tip' is a 3 µm tall pyramid with ~ 30 nm end radius.
- The electron-beam-deposited (EBD) tip or 'super tip' offers a higher aspect ratio (it is long and thin, good for probing pits) and sometimes a better end radius than the normal tip.
- The 'Ultra-lever' is based on an improved microlithography process.
- Ultra-lever offers a moderately high aspect ratio and on occasion a ~ 10 nm end radius.

Transmission Electron Microscopy (TEM)

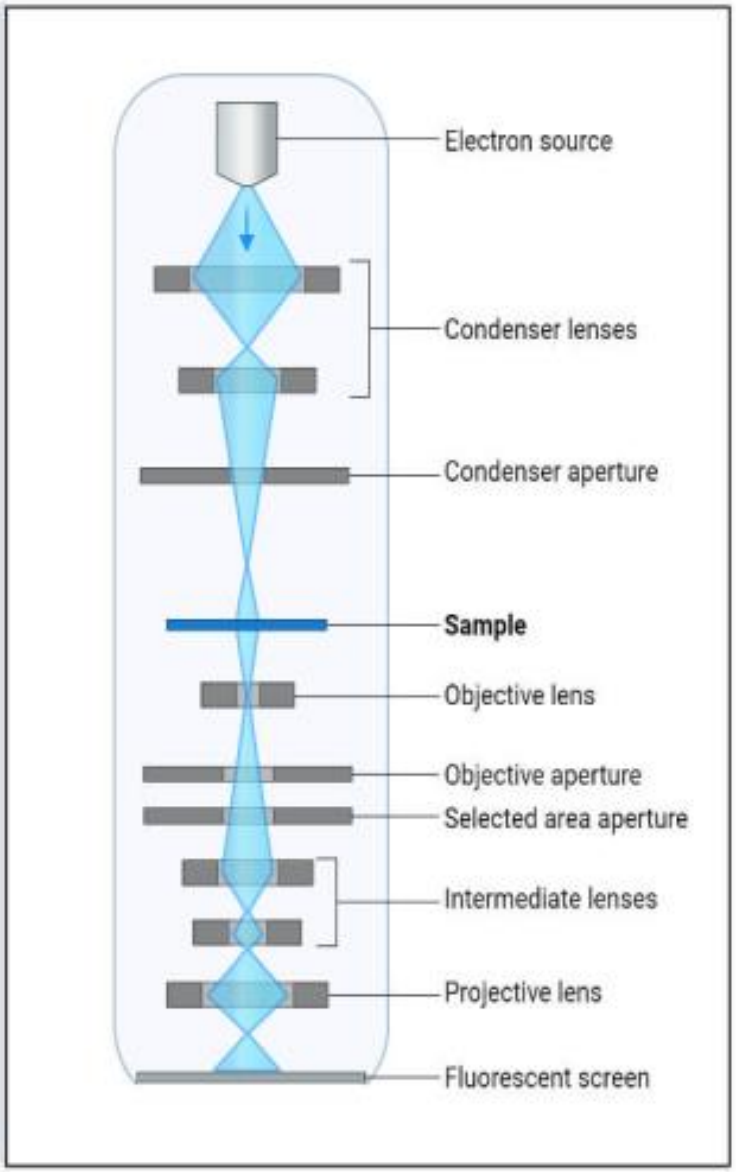
Principle:

- The working principle of the Transmission Electron Microscope (TEM) is similar to the optical microscope.
- The major difference is that optical microscopes use light rays to focus and produce an image while the TEM uses a beam of electrons to focus on the specimen, to produce an image.
- Electrons have a shorter wavelength in comparison to light photon which has a long wavelength.
- In optical microscopes decreases in the wavelength of the light increases the resolving power.
- The wavelength of the electrons is about 0.005nm which is 100,000 times shorter than that of light, hence TEM has better resolution than that of the optical microscope by about 1000times.
- Hence TEM can be used to detail the internal structures of the smallest objects like viruses.

Construction:

Transmission Electron Microscopy (TEM)





Electron gun:

- Electrons are produced by a cathode that is a heated V-shaped tungsten filament. The tungsten filament is covered by a control grid known as a Wehnelt cylinder made up of a central hole which lies columnar to the tube. The cathode and the control grid are negatively charged with an end of the anode which is disk-shaped that also has an axial hole.
- When electrons are transmitted from the cathode, they pass through the columnar aperture (hole) to the anode at high voltage with constant energy, which is efficient for focusing the specimen to produce an accurately defined image.
- It also has the condenser lens system which works to focus the electron beam on the specimen by controlling the energy intensity and the column hole of the electron gun. The TEM uses two condenser lenses to converge the beam of electrons to the specimen.
- The two condenser lens each function to produce an image i.e the first lens which has strong magnification, produces a smaller image of the specimen, to the second condenser lens, directing the image to the objectives.

Image- Producing system:

- Its made up of the objective lens, a movable stage or holding the specimen, intermediate and projector lenses. They function by focusing the passing electrons through the specimen forming a highly magnified image.
- The objective has a short focal length of about 1-5mm and it produces an intermediate image from the condenser which are transmitted to the projector lenses for magnification.
- The projector lenses are of two types, i.e the intermediate lens which allows great magnification of the image and the projector lens which gives a generally greater magnification over the intermediate lens.
- To produce efficient high standard images, the objectives and the projector lenses need high power supplies with high stability for the highest standard of resolution.

Image-Recording System:

- Its made up of the fluorescent screen used to view and to focus on the image. They also have a digital camera that permanently records the images captured after viewing.
- They have a vacuum system that prevents the bombardment or collision of electrons with air molecules disrupting their movement and ability to focus. A vacuumed system facilitates the straight movement of electrons to the image.
- The vacuume system is made up of a pump, gauge, valves and a power supply.
- The image that is formed is called a monochromatic image, which is greyish or black and white. The image must be visible to the human eye, and therefore, the electrons are allowed to pass through a fluorescent screen fixed at the base of the microscope.
- The image can also be captured digitally and displayed on a computer and stored in a JPEG or TIFF format. During the storage, the image can be manipulated from its monochromatic state to a colored image depending on the recording apparatus and use of pixel cameras can store the image in color.

Working of TEM:

- A heated tungsten filament in the electron gun produces electrons that get focus on the specimen by the condenser lenses.
- Magnetic lenses are used to focus the beam of electrons of the specimen. By the assistance offered by the column tube of the condenser lens into the vacuum creating a clear image, the vacuum allows electrons to produce a clear image without collision with any air molecules which may deflect them.
- On reaching the specimen, the specimen scatters the electrons focusing them on the magnetic lenses forming a large clear image, and if it passes through a fluorescent screen it forms a polychromatic image.
- The denser the specimen, the more the electrons are scattered forming a darker image because fewer electron reaches the screen for visualization while thinner, more transparent specimens appear brighter.